

A. Approach

The 3D reconstruction was created using a **Structure-from-Motion (SfM) photogrammetry pipeline** with 200 drone images captured in an orbital path around the lighthouse.

1. Image Preparation

The images were resized to a maximum of **1800 px** while keeping their GPS/EXIF data. This reduced memory usage on the CPU-only Kaggle environment.

2. Feature Extraction

SIFT was used to detect and describe important features such as corners, edges, and high contrast areas in each image.

3. Feature Matching

Since the drone followed a circular path, matching only neighboring images caused the reconstruction to break between the end and start of the flight. **Vocabulary tree matching** was therefore used to find similar images even when they were not captured next to each other.

4. Sparse 3D Reconstruction

COLMAP's **Incremental SfM** was used to estimate camera positions and create 3D points. Bundle adjustment was used to improve the camera poses and 3D points.

Results:

- **200/200 images registered**
- **147,301 3D points**
- **1.22 px mean reprojection error**

5. Dense Reconstruction

We attempted to create a denser 3D surface, However, it requires **CUDA/GPU support**, which was unavailable in the Kaggle CPU environment.

6. Depth Estimation

As a CPU-based demonstration, **OpenCV StereoSGBM** was used on two overlapping images to generate an example disparity/depth map.

B. Libraries and Tools

- **COLMAP** – Feature extraction, matching, SfM, and 3D reconstruction
- **OpenCV** – Stereo depth estimation
- **Open3D** – Point-cloud processing and visualization
- **Pillow (PIL)** – Image resizing and EXIF preservation
- **Kaggle Notebooks** – CPU-based computing environment

C. Limitations

- **Dense reconstruction was not completed** because the available COLMAP build did not support CUDA. OpenMVS was considered as a CPU based alternative but could not be implemented within the available time.
- **Camera trajectory drift occurred.** Although all 200 images were successfully registered, the camera path did not perfectly close because small pose errors accumulated.
- **The depth map is only illustrative.** It was generated from two unrectified images using StereoSGBM and is not part of the full SfM reconstruction. A production quality depth map would require multi-view stereo using the recovered camera poses.